

Introduction

Late-night salivary cortisol is a recommended first-line screening test for Cushing's syndrome. It is non-invasive, stress-free and offers high diagnostic sensitivity and specificity. As cortisol reference intervals (RI) are both method- and population-dependent, this study aimed to verify the manufacturer's RI for salivary cortisol in healthy Algerian adults according to the CLSI EP28-A3c guideline.

Methods

- Paired morning (7:00–9:00 AM) and late-night (midnight $\pm$ 30 min) saliva samples were collected by passive drool directly into sterile tubes from 20 healthy adults aged 18–65 years.
- Participants were instructed to refrain from eating, drinking, smoking, brushing their teeth for at least 2 h before sample collection.
 - Individuals with oral lesions or gingival bleeding were excluded.
 - Additional exclusion criteria included :
 - BMI ≥ 25 kg/m², pregnancy, chronic diseases (diabetes mellitus, hypertension, renal, cardiac, or hepatic disease), acute infection, night-shift work, and the use of exogenous glucocorticoids or drugs that could interfere with the hypothalamic–pituitary–adrenal axis.
- Salivary cortisol was measured by electrochemiluminescence immunoassay (Elecsys Cortisol II, Roche Diagnostics).
- Statistical analyses were performed using RStudio. Data distribution was assessed using the Shapiro–Wilk test.
- Outliers were identified using Tukey's method.
- The manufacturer's RI was considered verified if no more than 2/20 ($\leq 10\%$) results fell outside the reference limits (Figure 1).

Results

Twenty healthy adults (16 females and 4 males) aged between 22 and 64 years (mean age: 34 years) were included in the analysis. Baseline characteristics of subjects are presented in Table 1. Paired salivary cortisol concentrations measured in the morning and at midnight are shown in Figure 2.

- Mean ($\pm$ SD) morning salivary cortisol was **7,87 $\pm$ 2,94 nmol/L** (Figures 3 and 4).
- Median (range) midnight salivary cortisol was **2,11 nmol/L** (1,5–5,25), 45% of midnight values were below the assay's limit of detection (LOD, $<1,5$ nmol/L).
- Two morning values (16,2 and 17,8 nmol/L) were identified as outliers by Tukey's method.
- These values were excluded and replaced with two new samples.
- All morning and late-night salivary cortisol values were below the manufacturer's upper reference limits (morning: $<24,1$ nmol/L; late-night: $<11,3$ nmol/L).**

Table 1. Characteristics of the study population

Characteristic	Healthy adults (n = 20)
Age, years	34.2 $\pm$ 12.2
Female, n (%)	16 (80.0)
Male, n (%)	4 (20.0)
BMI, kg/m ²	22.6 $\pm$ 2.6

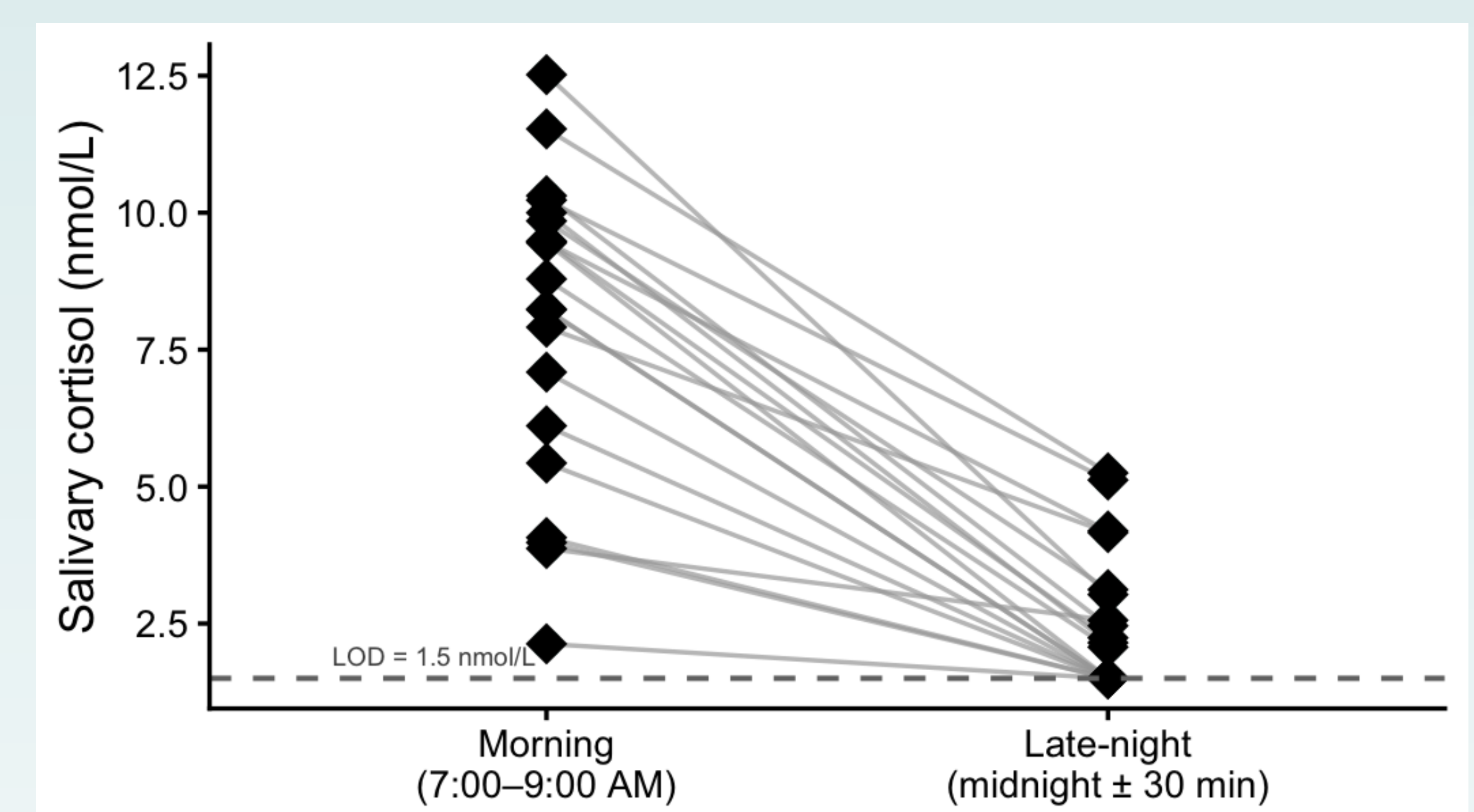


Figure 2. Paired Morning and Late-Night Salivary Cortisol in Healthy Adults

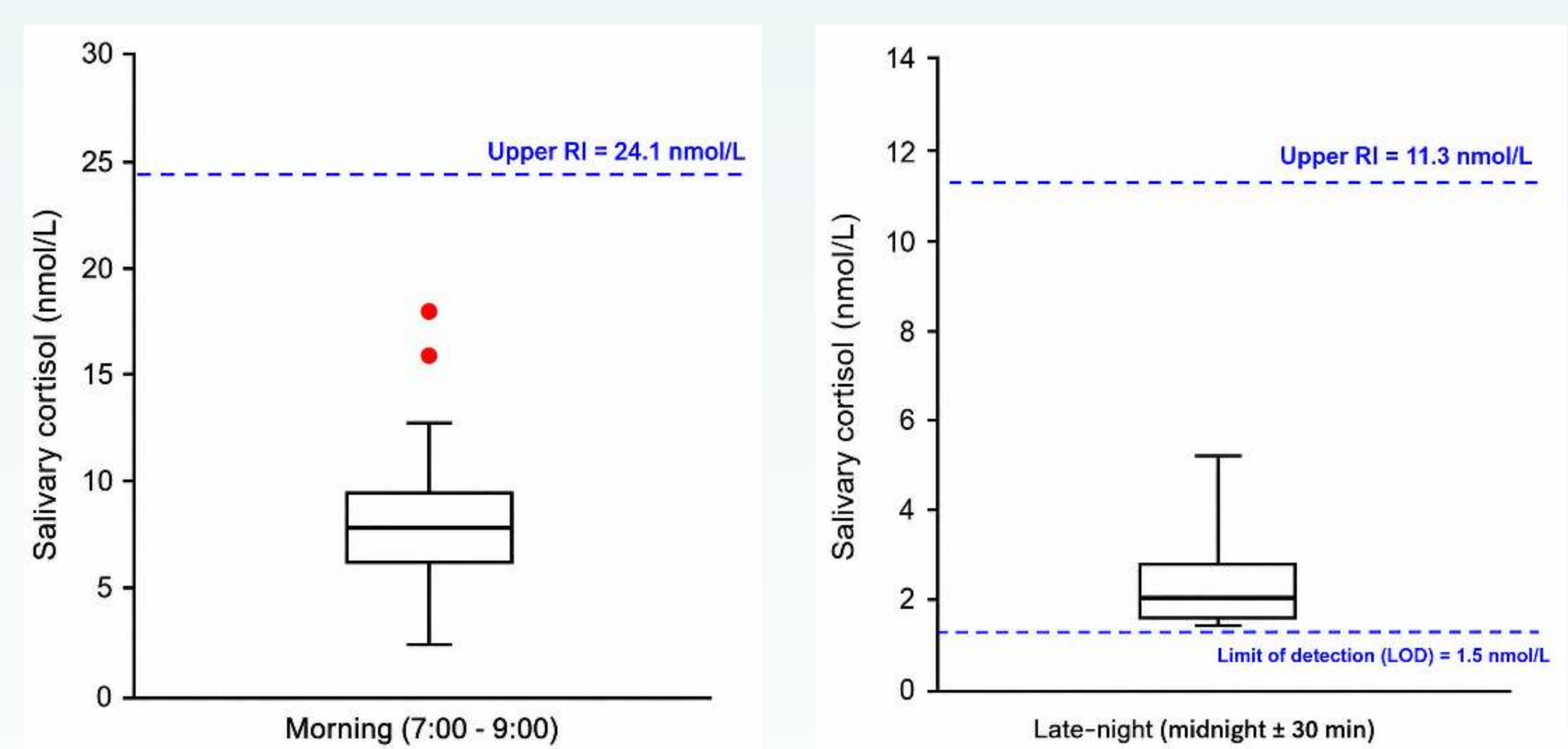


Figure 3. Mean and median salivary cortisol concentrations at morning and late night

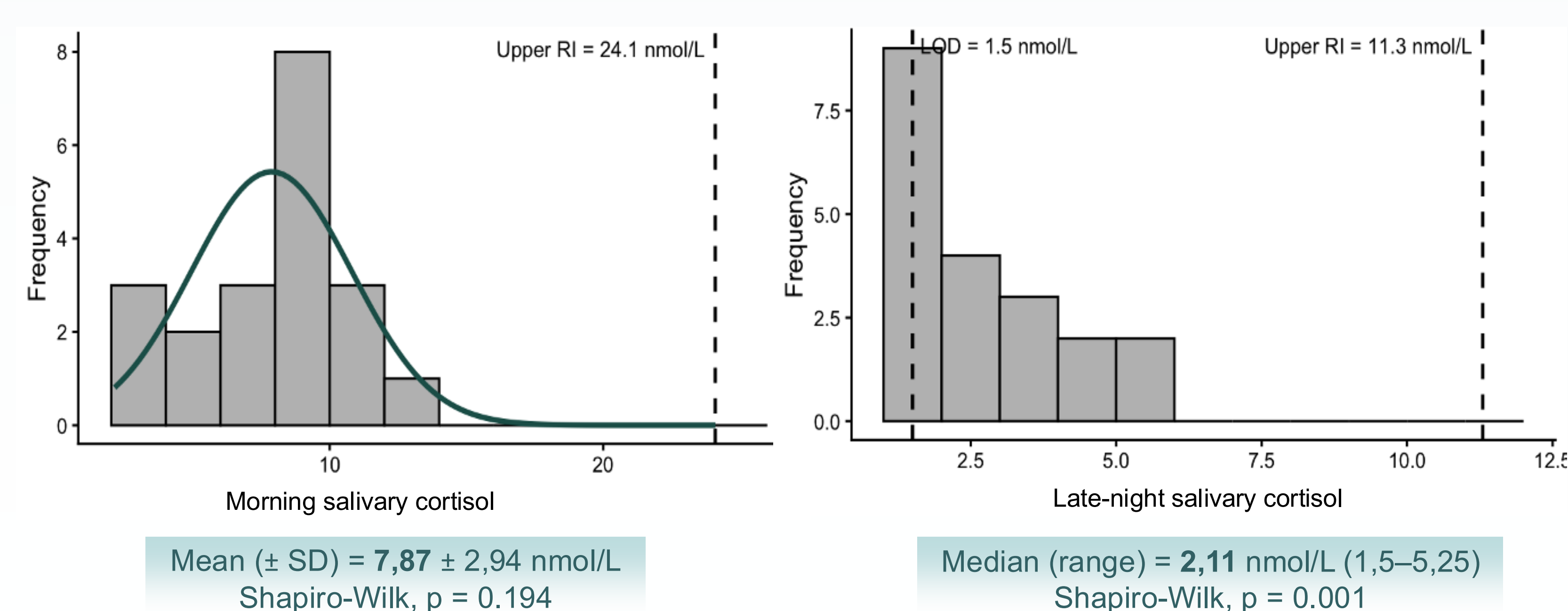


Figure 4. Distribution of morning and late-night salivary cortisol concentrations

Figure 1. Verification of reference intervals according to the CLSI EP28-A3c approach. Adapted from Ozarda et al. (Clin Chem Lab Med. 2019;57:30–37)

Conclusion

The manufacturer's reference intervals for morning and late-night salivary cortisol (Elecsys Cortisol II) were verified in this healthy Algerian adult population, supporting their direct clinical use for Cushing's syndrome screening in our laboratory.